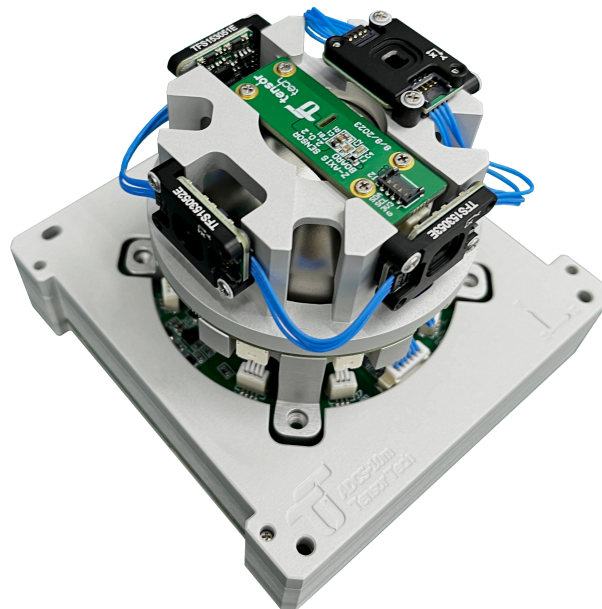




User Manual for Integrated Attitude Determination and Control System: ADCS-MTQ, ADCS-10m, ADCS-20m, and ADCS-40m



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List of Abbreviations

All abbreviations are capital letters. Bold font is applied only for the initial use.

ADCS	Satellite Attitude Determination and Control System
ADCS-MCB	ADCS Main Control Board
ADCS-x	ADCS-x refers to any product within the ADCS series, including ADCS-MTQ, ADCS-10m, ADCS-20m, and ADCS-40m
COM	Center of Mass
CPT	Comprehensive Performance Test
DUT	Device Under Test
ECEF	Earth-Centered-Earth-Fixed
ECI	Earth-Centered Inertial
EM	Engineering Model
EPS	Electrical Power System
ESD	Electrostatic Discharge
FOV	Field Of View
FSS	Fine Sun Sensor
FSM	Finite State Machine
FM	Flight Model
GEVS	General Environmental Verification Standard
HW	Hardware
IMU	Inertial Measurement Unit
LEO	Low Earth Orbit
LPT	Limited Performance Test
LSB	Least Significant Bit
LVLH	Local-Vertical-Local-Horizontal
MRST	Master Receive Slave Transmit
MSB	Most Significant Bit
MTQ	Magnetic Torquer
MTSR	Master Transmit Slave Receive
MPN	Manufacturer Part Number
NVM	Non-Volatile Memory
OBC	Onboard Computer on a satellite
PCBA	Printed Circuit Board (PCB) Assembly

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PCB	Printed Circuit Board
PTFE	Polytetrafluoroethylene
QM	Qualification Model
RPM	Revolution per Minute
RTC	Real-Time Clock
SEE	Single Event Effect
SMT	Surface Mount Technology for PCBA
SWaP	Size
TID	Total Ionizing Dose radiation test
TLE	Two-Line Element set
TVAC	Thermal Vacuum Chamber
VSCMG	Variable-speed Control Moment Gyroscope

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Nomenclature

$N = \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$	The ECI frame, where $\hat{n}_1$, $\hat{n}_2$, and $\hat{n}_3$ are the basis vectors
$B = \{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$	The ADCS frame. where $\hat{b}_1$, $\hat{b}_2$, and $\hat{b}_3$ are the basis vectors
$H = \{\hat{h}_1, \hat{h}_2, \hat{h}_3\}$	The satellite frame. $\hat{h}_1$, $\hat{h}_2$, and $\hat{h}_3$ are the basis vectors
$O = \{\hat{o}_1, \hat{o}_2, \hat{o}_3\}$	The LVLH frame. $\hat{o}_1$, $\hat{o}_2$, and $\hat{o}_3$ are the basis vectors
$S = \{\hat{s}_1, \hat{s}_2, \hat{s}_3\}$	The sensor frame. $\hat{b}_1$, $\hat{b}_2$, and $\hat{b}_3$ are the basis vectors
$G = \{\hat{g}_s, \hat{g}_t, \hat{g}_g\}$	The gimbal frame. $\hat{g}_s$, $\hat{g}_t$, and $\hat{g}_g$ are the basis vectors
${}^N\mathbf{r}_{B/N}$	3×1 position vector of the ADCS (B) relative to the earth center (N) in the ECI frame
${}^N\mathbf{r}_{T/N}$	3×1 position vector of the target (T) relative to the earth center (N) in the ECI frame
${}^N\mathbf{r}_{T/B}$	3×1 position vector of the target (T) relative to the ADCS (B) in the ECI frame
${}^B\mathbf{T}_N$	3×3 transformation matrix from the ECI frame to the ADCS frame
${}^B\hat{\mathbf{p}}$	3×1 body fixed vector which is desired axis to point the target
$\mathbf{q}$	4×1 quaternion vector
$\hat{\mathbf{e}}$	3×1 principle rotation axis
ψ	the principle rotation angle
θ	Polar angle of incident light of the FSS
ϕ	Azimuthal angle of incident light of the FSS
γ	The gimbal angle of the VSCMG
Ω	The wheel speed of the VSCMG

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1. INTRODUCTION

Tensor Tech's integrated nano-satellite **Attitude Determination and Control Subsystem (ADCS)** is introduced in this user manual. Tensor Tech's ADCS is composed of hardware and firmware. The former consists of actuators, sensors, and a dedicated computer; the latter provides the basic and advanced automatic modes for the users to control the attitude of their satellite.

This user manual document is aimed at providing Tensor Tech ADCS users with all the required necessary information for **Assembly, Integration, and Test (AIT)** activities as well as) for sending commands and accessing states of the ADCS to operate the system and, therefore, the attitude of the satellite in Space. By familiarizing with the contents of this manual, the user should be able to handle and operate Tensor Tech's ADCS system.

For the technical specifications of each of the subsystems, please refer to their datasheet and User Manuals; for the mechanical and electrical interface, please refer to the respective **Interface Control Document (ICD)**.

2. UNPACKING AND HANDLING

2.1 Guideline

1. Control the Electrostatic Discharge (ESD) in the environment

ADCS-x is an ESD-sensitive device. Please unpack and integrate with the proper equipment. The procedures mentioned in this document must be strictly followed.

2. Control the contamination in the environment

ADCS-x is assembled in the cleanroom environment certified with ISO 14644-1 and ISO 7 standard. It is majorly preventing dust from getting into the ADCS-x. To achieve the proper integration of ADCS-x with your satellite, it is suggested to conduct in the cleanroom environment with ISO 7 standard, or at least under an air shower with ISO 7 standard to avoid contamination from other systems before launching into space.



3. Remove the Kapton Tape before launch

Kapton tape is applied to cover the top of the lens for additional protection from contamination. This tape is required to be removed after the final integration of the satellite onto the rocket for FSS-15/FSS-15M/FSS-15D to operate as intended. If users plan to do functional tests with FSS-15/FSS-15M/FSS-15D, for instance under a solar simulator, the applied piece of Kapton tape is required to be removed during the test, and then apply a new piece of Kapton tape after the test.

4. Do not remove any electrical or mechanical components on the ADCS-10m

Most of the **Surface-Mount-Technology (SMT)** electrical components and screws are glued with space-grade epoxy before the shipment of ADCS-x to prevent any possible fall-off situation happening during launching or vibration tests. Please contact Tensor Tech team if any fall-off situation is observed after the unpacking or similar environmental tests which are mentioned in this manual.

2.2 Packing Lists

Users can find below the list of components delivered with the corresponding ADCS-Series configuration. Please refer to the ordered configuration.

2.2.1 ADCS-MTQ Suite

Table 2-1. Packing List for Tensor Tech's ADCS-MTQ Suite

Name	Quantity	Description
ADCS-MCB with MTQ	1	Main control board for ADCS-MTQ with 2 MTQ-200m and 1 air coil integrated
FSS-15M	1	Refer to the cable specifications provided under the " Polytetrafluoroethylene (PTFE) connection cable" description. Such cable is required for connecting the FSS-15M to the ADCS-MCB.
PTFE Connection Cable	-	PTFE cable connects the 4-pin pico-EZmate header on the FSS-15M or FSS-15 to the 4-pin pico-blade header on the ADCS-MTQ or 4-pin pico-EZmate header on other FSS-15 to form a daisy-chain. The user is required to specify the length and quantity of these cables.
FSS-15	¹ Optional	PTEE connection cable is required to connect this FSS-15 to the ADCS-MCB.

2.2.2 ADCS-10m Suite

Table 2-2. Packing List for Tensor Tech's ADCS-10m Suite

Name	Quantity	Description
ADCS-MCB with MTQ	1	Main control board for ADCS-10m with 2 MTQ-200m torquers and 1 air coil integrated
CMG-10m	1	Major actuator, Control Moment Gyro
FSS-15	5	Note: PTFE connection cable is required to connect FSS-15 to the ADCS-MCB. <u>Configuration 1 (check figure 4-2 in ADCS Series Datasheet for details):</u> Five or fewer FSS-15 will be pre-installed on the CMG-10m, and cables will be wired up also. The rest of the FSS-15 should be installed on the satellite by the user. <u>Configuration 2 (check figure 4-2 in ADCS Series Datasheet for details):</u> None of the FSS-15 will be pre-installed on the

¹ Although the standard offer of ADCS-MTQ comes with one FSS-15M only, the customers still have the option to connect multiple FSS-15 to ADCS-MCB for broader sky coverage and redundancy.

Name	Quantity	Description
		CMG-10m; Tensor Tech delivers standard-length cables for them. The user may prepare the PTFE cable by themselves to hook these five FSS-15 up if the standard cable does not meet their requirement.
FSS-15M	1	Note: PTFE connection cable is required to connect FSS-15M to the ADCS-MCB.
PTFE Connection Cables	-	Standard PTFE cables connect the 4-pin pico-EZmate header on the FSS-15 or FSS-15M to the 4-pin pico-blade header on the ADCS-MCB or 4-pin pico-EZmate header on other FSS-15 to form a daisy-chain. Standard cables for wiring ADCS-MCB to CMG-10m.

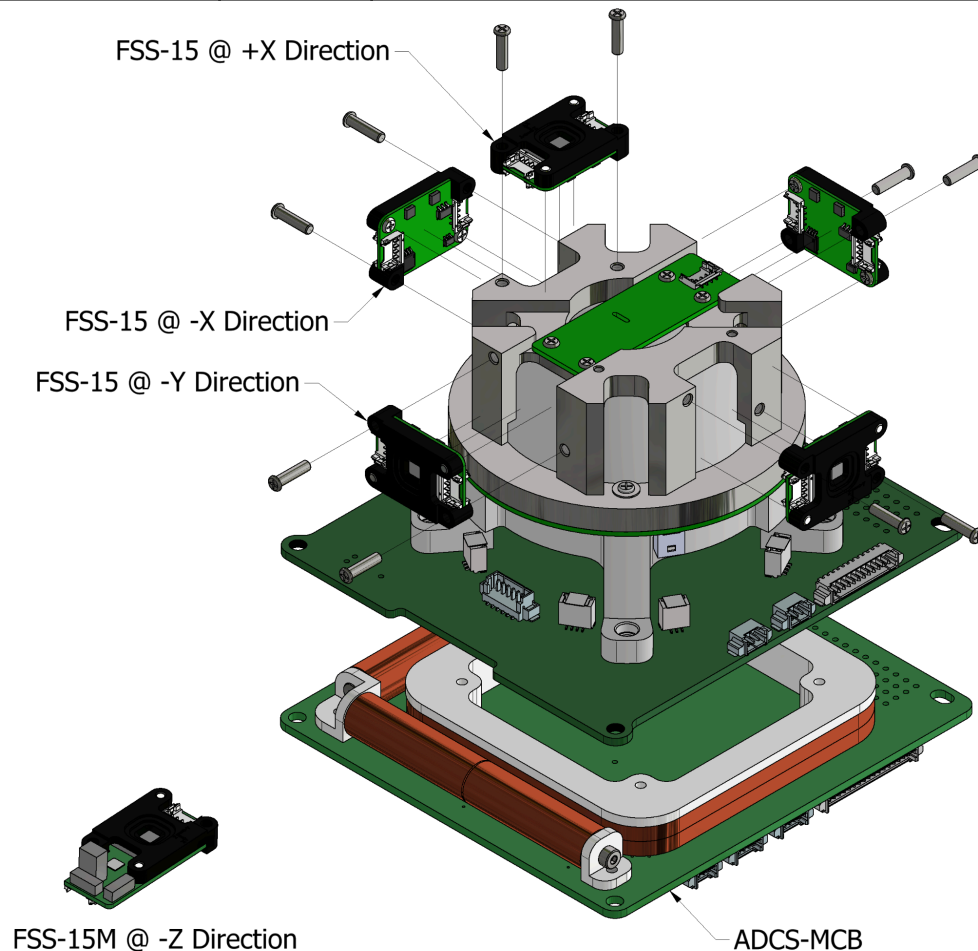


Figure 2-1. Hardware Assembly of ADCS-10m

2.2.3 ADCS-20m Suite

Table 2-3. Packing List for Tensor Tech's ADCS-20m Suite

Name	Quantity	Description
ADCS-MCB with MTQ	1	Main control board for ADCS-20m with 2 MTQ-200m torquers and 1 air coil integrated.
CMG-10m	2	Major actuator, Control Moment Gyro
FSS-15	5	Note: PTFE connection cable is required to connect FSS-15 to the ADCS-MCB. <u>Configuration 1 (check figure 4-2 in ADCS Series Datasheet for details):</u> Five or fewer FSS-15 will be pre-installed on the 2 CMG-10m, and cables will be wired up also. The rest of the FSS-15 should be installed on the satellite by the user. <u>Configuration 2 (check figure 4-2 in ADCS Series Datasheet for details):</u> None of the FSS-15 will be pre-installed on the CMG-10m; Tensor Tech delivers standard-length cables for them. The user may prepare the PTFE cable by themselves to hook these five FSS-15 up if the standard cable does not meet their requirement.
FSS-15M	1	Note: PTFE connection cable is required to connect FSS-15M to the ADCS-MCB.
PTFE Connection Cables	-	Standard PTFE cables connect the 4-pin pico-EZmate header on the FSS-15 or FSS-15M to the 4-pin pico-blade header on the ADCS-MCB or 4-pin pico-EZmate header on other FSS-15 to form a daisy-chain. Standard cables for wiring ADCS-MCB to CMG-10m.

2.2.4 ADCS-40m Suite

Table 2-4. Packing List for Tensor Tech's ADCS-40m Suite

Name	Quantity	Description
ADCS-MCB	1	Main control board for ADCS-40m.
MTQ-2	3	Torquer rod designed for >12U nano-satellite and microsatellite.
CMG-10m	4	Major actuator, Control Moment Gyro
FSS-15	5	Note: PTFE connection cable is required to connect FSS-15 to the ADCS-MCB. <u>Configuration 1 (check figure 4-2 in ADCS Series Datasheet for details):</u> Five or fewer FSS-15 will be pre-installed on the 4 CMG-10m, and cables will be wired up also. The rest of the FSS-15 should be installed on the satellite by the user. <u>Configuration 2 (check figure 4-2 in ADCS Series Datasheet for details):</u> None of the FSS-15 will be pre-installed on the CMG-10m; Tensor Tech delivers standard-length cables for them. The user may prepare the PTFE cable by themselves to hook these five FSS-15 up if the standard cable does not meet their requirement.
FSS-15M	1	Note: PTFE connection cable is required to connect FSS-15M to the ADCS-MCB.
PTFE Connection Cables	-	Standard PTFE cables connect the 4-pin pico-EZmate header on the FSS-15 or FSS-15M to the 4-pin pico-blade header on the ADCS-MCB or 4-pin pico-EZmate header on other FSS-15 to form a daisy-chain. Standard cables for wiring ADCS-MCB to CMG-10m.

3. STORAGE

Tensor Tech's products are stored in a relative humidity 2% environment in the factory. It is recommended to store in a moisture-proof box with a relatively low humidity environment. (20% as the highest tolerable humidity)

4. SYSTEM OVERVIEW

As Figure 4-1 shows, ADCS-x comprises four parts: "Register Map," "Finite State Machine (FSM)," "Hardware (HW) Peripheral," and "Actuators/Sensors." Here is the detailed description of each part:

1. Register Map: the interface where the user exchanges data, such as status flag, reference quaternion, sensor measurements, etc. ADCS-x provides UART and I²C to read/write the register map.
2. FSM: performs the attitude determination and control algorithm according to the settings inside the register map.
3. HW Peripheral: offload the FSM computation effort by buffering the sensors' data and modulating the control signal.
4. Actuators/Sensors: ADCS-x has 0/1/2/4 CMG-10m and three orthogonal magnetorquer rods, serving as the actuators in the control system. Six fine sun sensors, one 3-axis magnetometer, and one IMU are the primary sensors utilized in ADCS-x. The GNSS module is recommended to achieve better pointing results.

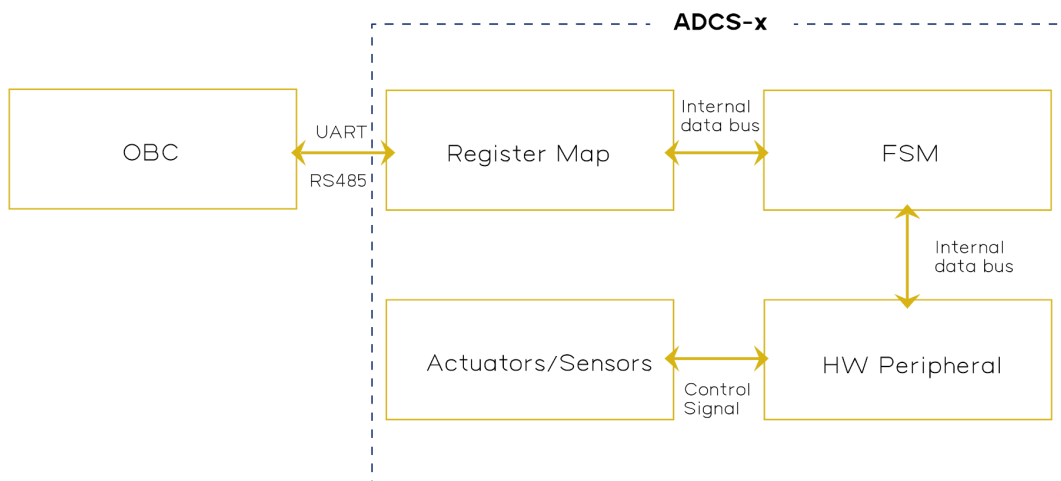


Figure 4-1. ADCS-x System Block Diagram

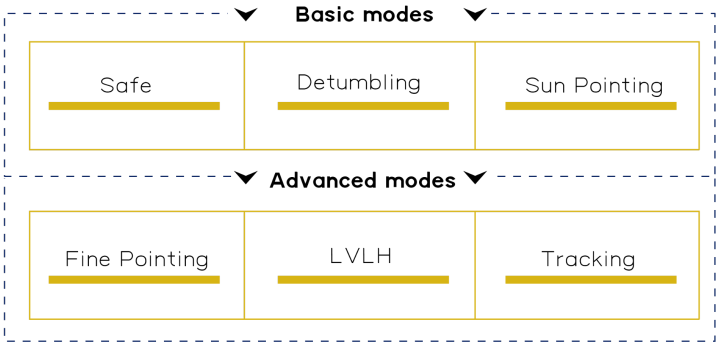


Figure 4-2. ADCS-x Control Modes

As Figure 4-2 shows, ADCS-x is embedded with two types of modes, “basic mode” and “advanced mode.” Users can switch between three basic modes without preconditions. The advanced mode requires users to update orbital information, current time, and TLE periodically. It is recommended to update the TLE every time the satellite contacts the ground station. If the TLE isn’t updated for more than 48 hours in LEO orbit, the attitude determination quality cannot be guaranteed. Moreover, the GNSS information can serve as additional information that makes the satellite position and time prediction more accurate.

The user can switch between any of the two modes. However, as figure 4-3 shows, ADCS-x mandatorily switches back to Safe Mode or De-tumbling Mode under certain conditions. It is recommended that OBC monitor the status flags and current of the ADCS-x every 1 to 5 minutes.

Please note that ADCS-MTQ configuration only supports Basic Mode, while ADCS-10m, ADCS-20m-Discrete and ADCS-40m-Discrete support both basic mode and advanced Mode. The register related to advanced Mode in ADCS-MTQ configuration is read-only or unimplemented.

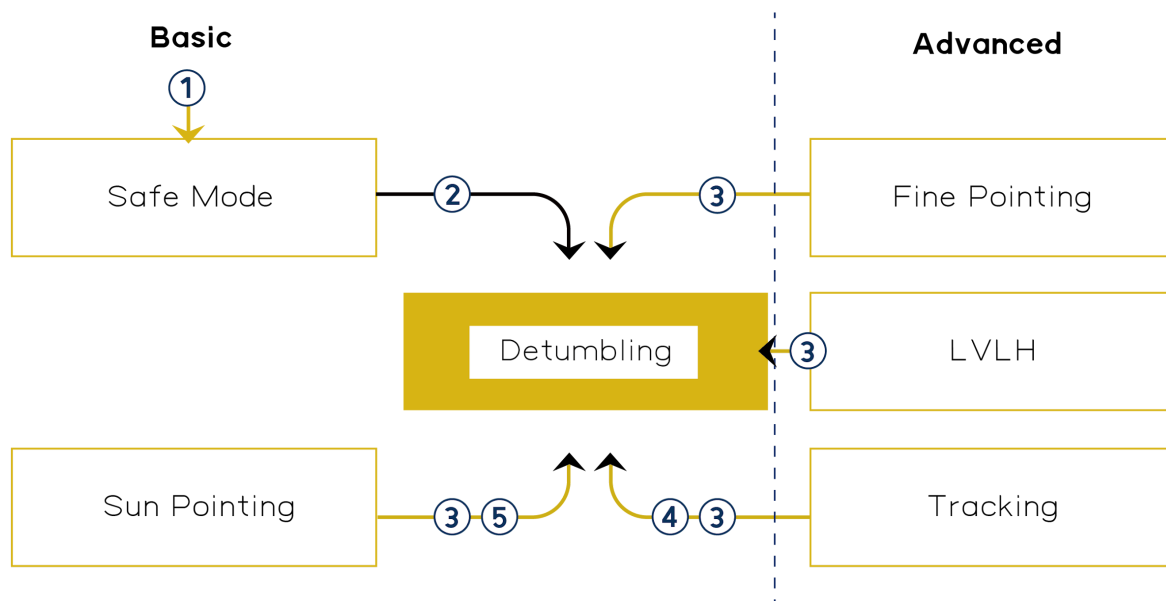


Figure 4-3. Mandatory mode transition diagram

- Condition ①: Soft error, hard error, or tumbling rate > 360 deg/second
The events that will be deemed as soft errors include
 - 1) over-temperature protection (any one of the temperature sensors on ADCS-x detects a temperature that is higher than its absolute operating temperature maximum)
 - 2) over-current protection (any one of the 3.3V and 5V power buses exceed its absolute maximum power limit, its error flag status OC=1)
 - 3) under-voltage protection (any one of the 3.3V and 5V power buses exceed its absolute maximum supplied voltage range, its error flag status UV=1)
 - 4) hardware reset (watching dog time-out, its error flag status WDT=1)

The ADCS-x mandatorily switches back to Safe Mode whenever a soft error occurs, or the satellite's tumbling rate is larger than 360 deg/second.

- Condition ②: Tumbling rate between 5 deg/second to 360 deg/second and rotor speed > 1000 rpm.
To avoid excess angular momentum stored in the ADCS-x during Safe Mode, ADCS-x switches to De-tumbling Mode and gradually dumps the angular momentum.
- Condition ③: Tumbling rate > 5 deg/second
Except for the Safe Mode, ADCS-x switches to De-tumbling Mode when the satellite starts tumbling.
- Condition ④: Target lost
The elevation angle of the target object exceeding the predefined threshold, making it no longer feasible to maintain a lock.
- Condition ⑤: Sun vector invalid
When the Sun light is not captured by any FSS and the user does not manually input a sun vector, ADCS-x in the Sun Pointing Mode switches to De-tumbling Mode.

Please note that the above five conditions are the only conditions between which ADCS-x actively switches mode. Otherwise, ADCS-x remains in its previously selected mode.

The impacts and definitions of the soft error have been described in Condition ①. On the other hand, the definition of hard error is that it will cause permanent consequences on the component, and it rarely recovers by itself, e.g., destruction of the memory, shortage of the circuit, or non-functioning of the driving circuit. Therefore, we restrict the user from some control modes based on the type of hard error, which will be described below.

However, the ADCS-x will continually detect each component's health status every 100ms. Once such components recover, the originally banned modes can be restarted. A restart of power may help to recover the hard error.

The impacts of hard errors on attitude sensors are as follows:

- 1) All the advanced modes can not be triggered if the IMU has a hard error.
- 2) If all FSS-15s fail to retrieve sun vectors, both the advanced mode and Sun-pointing Mode cannot be triggered.
- 3) If the magnetometer has a hard error, the system will mandatorily enter Safe Mode.

Moreover, the impacts of hard error attitude actuators are listed below:

- 1) If every CMG has a hard error, all the advanced modes can not be triggered.
- 2) If the magnetorquer has a hard error, the system will lock at Safe Mode.

When the user tries to switch the ADCS-x to a mode that violates the above rules due to hard error, such a selected mode will not be able to trigger. Under such

circumstances, the ADCS-x will automatically be switched to Safe Mode.

The hard error conditions for both the attitude sensor and actuators are listed in table 4-2 and 4-3.

Table 4-2. Sensor error that will be deemed as a hard error (SEN_ERR)

Component	Status of its error flag	Description
3-axis Magnetometer	All MAGx =1, x=0,1,2,3,4,5	ADCS computer fails to communicate with all magnetometers
FSS-15 Fine Sun Sensor	All FSSx =1, x=0,1,2,3,4,5	ADCS computer fails to communicate with all fine sun sensors
Inertial Measurement Unit (IMU)	All IMUx =1, x=0,1,2,3	ADCS computer fails to communicate with all IMUs
Star Tracker	All STRx =1, x=0,1	ADCS computer fails to communicate with all star trackers

Table 4-3. Actuator error that will be deemed as a hard error (ACT_ERR)

Component	Status of its error flag	Description
CMG-10m CMG	All CMGx =1, x=0,1,2,3	ADCS computer fails to communicate with all CMG-10m or short-circuit on any phase circuitry of the CMG-10m is sensed
3-axis Magnetorquer	All MTQx =1, x=0,1,2	Short-circuit or open-circuit on all magnetorquer is sensed

5. CONTROL MODES

This chapter introduces the control modes of the ADCS-x. If no mandatory event trigger, the user is allowed to switch all control modes by altering the **MODE** register setting.

5.1 Safe Mode

Safe Mode is the initial state of ADCS-x when the EPS powers it up. ADCS-x halts all actuators in Safe Mode but keeps monitoring and updating the system status. The output result is stored in the register map. ADCS-x consumes the lowest power in Safe Mode, and no attitude control is executed.

- **Trigger:** The user manually sets the MODE bits in CONF or the mandatory condition 1 occurs, ADCS-x enters Safe Mode.
- **Task:** ADCS-x measures all sensor and actuator states, e.g. magnetometer, FSS and IMU measurements; VSCMG wheel rate; temperatures, and power consumption, etc.
- **Output:** The measured results are stored inside the register map, including
 - 1) STAT: hard, soft, and others sensor status.
 - 2) ACT_ERR, SEN_ERR, CSS_ERR: actuators, sensors, and CSS status.
 - 3) Q0/Q1/Q2/Q3, RATEX/RATEY/RATEZ: attitude and attitude rate determination.
 - 4) LAT/LONG/ALT: satellite position in the geodetic coordinates.
 - 5) ADCS_TMP, CMGx_TMP, FSS_TMP1, FSS_TMP2, FSS_TMP3: all devices temperature.
 - 6) SVX/SVY/SVZ: sun vector.
 - 7) MAGX/MAGY/MAGZ: magnetometer measurement.

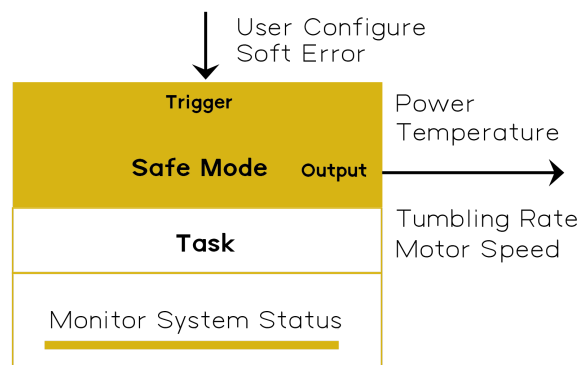


Figure 5-1. Safe Mode

5.2 De-tumbling Mode

If the detected satellite tumbling rate is larger than 360 deg/second², the de-tumbling control algorithms and actuators (3-axis magnetorquers) will not be triggered and will automatically enter the Safe Mode. However, the ADCS will continuously estimate the satellite tumbling rate in the Safe Mode. Once the OBC finds that the tumbling rate gets smaller than 360 deg/second, it can switch the ADCS back to the De-tumbling Mode.

The B-dot-based de-tumbling function will be activated in the De-tumbling Mode. It can de-tumbling the satellite from 30 deg/second to 1 deg/second will take less than 12 hours.

- **Trigger:** To enter De-tumbling Mode, the user manually sets the MODE bits in CONF or mandatory condition 2 occurs in the Safe mode or mandatory condition 3 occurs in the rest of states.
- **Task:** ADCS-x monitors system status and updates the corresponding part of the register map. The actuators exert torque to slow down the satellite's tumbling rate.
- **Input:** Optionally, OBC can feed in TLE or GNSS information; the ADCS-x will then perform attitude determination.
- **Output:** The system status registers, power, and temperature included in Safe Mode are updated. If OBC provides TLE or GNSS information, ADCS-x generates the estimated attitude.

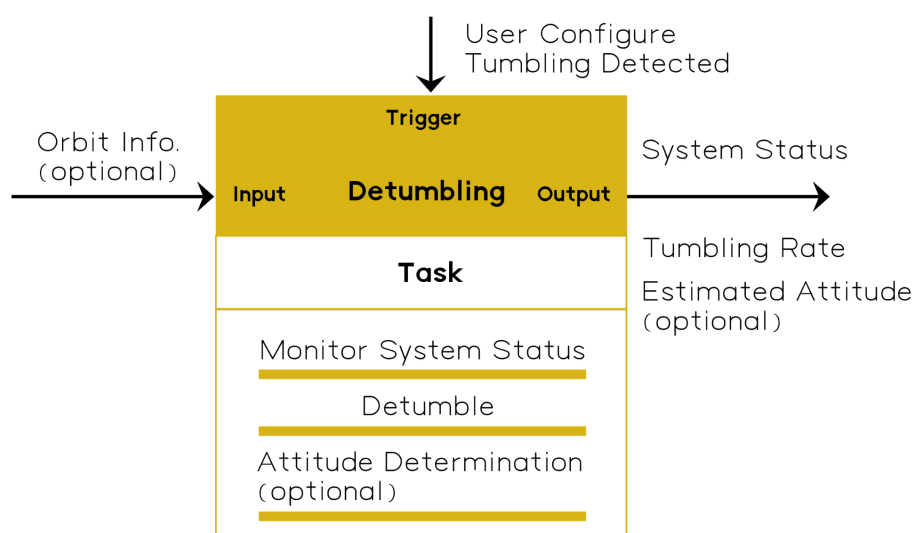


Figure 5-2. Flowchart of De-tumbling Mode

² Such a high tumbling rate is unlikely to happen unless there is some anomaly on the deployer or rocket after the launch. The malfunction of the thrusters or a strike by space debris may have the chance to cause such a tumbling rate as well. When such a high tumbling rate event happens, the OBC should keep the ADCS to Safe Mode and wait for a certain period for the satellite to decrease its tumbling rate. Such a passive reduction on the satellite's tumbling rate is feasible by using the magnetic hysteresis effect that will dissipate the satellite's kinetic energy during such a high tumbling rate. However, the rate for such a tumbling rate reduction depends mainly on the material used on the satellite. After the tumbling rate goes lower than 360 deg/second, the OBC can then switch the ADCS to De-tumbling Mode using 3-axis magnetorquers.

5.3 Sun Pointing Mode

The user has to specify a unit vector in the Satellite Frame that is parallel to the normal vector of the face of the satellite that is going to be pointed to the sun. Following the mission launch, the user can monitor the status of individual FSS-15 units from the SEN_ERR register. Unlike our other products that support three-axis stabilization, the "ADCS-MTQ" is designed exclusively for spin stabilization, operating without CMGs. The effectiveness of "ADCS-MTQ" is largely influenced by the selection of the stabilization axis and the satellite's mass distribution. Choose the axis alignment carefully to optimize stabilization performance.

- **Trigger:** The user manually sets the MODE bits in CONF.
- **Task:** ADCS-x monitors system status and updates the register map. When the sun appears in the field of view, and the satellite is not in eclipse, ADCS-x keeps the configured vector, **POINTING_AXIS_X/Y/Z**, pointing to the sun.
- **Input:** Optionally, OBC can feed in TLE or GNSS information³.
- **Output:** In every control cycle, the system status e.g. tumbling rate, sun vector, and estimated attitude, etc. are updated from [User\(0\)](#) register map, while sensor and actuator raw data are updated from [Sensor/Actuator\(1\)](#) register map.

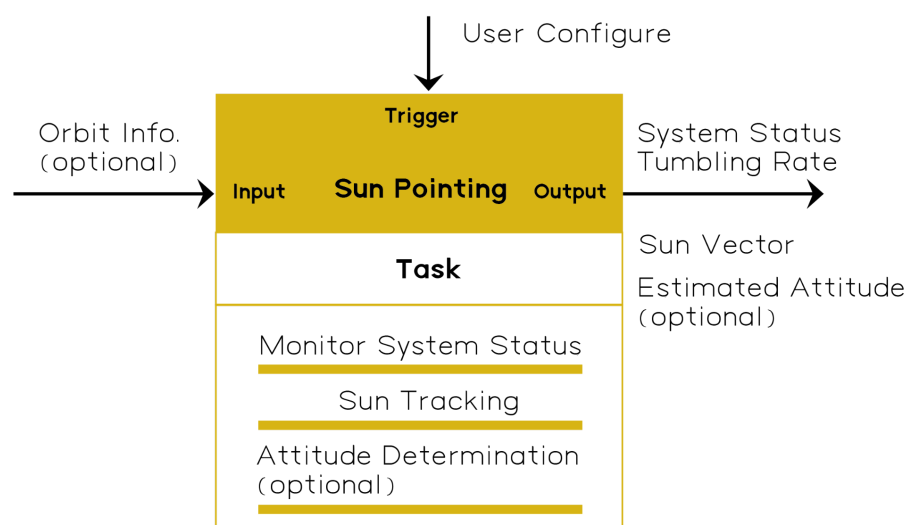


Figure 5-3. Flowchart of Sun-pointing Mode

³ In the case of entry-level products of ADCS such as the ADCS-MTQ, the availability of time and position information becomes critical when the presence of equipped sun sensors is limited. Consequently, the inclusion of TLE or GNSS information becomes essential to compensate for the deficiency in sun sensor inputs.

5.4 Fine Pointing Mode

The user can input the reference attitude in quaternion to the ADCS-x, and the ADCS-x will rotate the satellite accordingly. Please note that the slew rate between commands should be controlled within 5 degrees/second. If exceeding such a limit, the satellite might not respond to the command accurately. Moreover, when the ADCS detects a tumbling rate that is higher than 5 deg/second, it will switch to De-tumbling Mode automatically.

In this operation mode, users are granted the flexibility to specify any attitude command relative to the Earth-Centered Inertial (ECI) frames, particularly focusing on the Geocentric Celestial Reference Frame (GCRF). This specific frame is one of several within the ECI system, offering a precise reference for orienting the Earth satellite.

This mode utilizes inertial pointing control to maintain a consistent orientation relative to the fixed stars, ensuring zero angular velocity. Users can create and implement custom time-based attitude profiles. It's crucial, however, to design these changes in attitude to be gradual enough to allow the hardware to adjust smoothly and accurately between commands.

Moreover, the bandwidth of the ADCS-x is 5Hz, and the user can send their command at any update rate that they desire. However, the internal attitude control loop will down-sample the update rate. Therefore, there's no need to update the attitude command >10Hz in Fine Pointing Mode.

- **Trigger:** The user manually sets the MODE bits in CONF.
- **Task:** ADCS-x monitor system status and update register map. Given the desired attitude quaternion, ADCS-x sets that quaternion as a reference in the control loop.

Parameter	Q0REF	Q1REF	Q2REF	Q3REF
Value	1	0	0	0

Table: Reference Command Example for Fine Pointing Mode

By default, the system uses a unit quaternion, aligning the ADCS frame's X, Y, and Z axes directly with those of the GCRF.

- **Input:** TLE plus current time or GNSS information is required. When sending the commands, the attitude quaternion should be given by the OBC.
- **Output:** In every control cycle, the system status e.g. tumbling rate, sun vector, and estimated attitude, etc. are updated from [User\(0\)](#) register map, while sensor and actuator raw data are updated from [Sensor/Actuator\(1\)](#) register map.

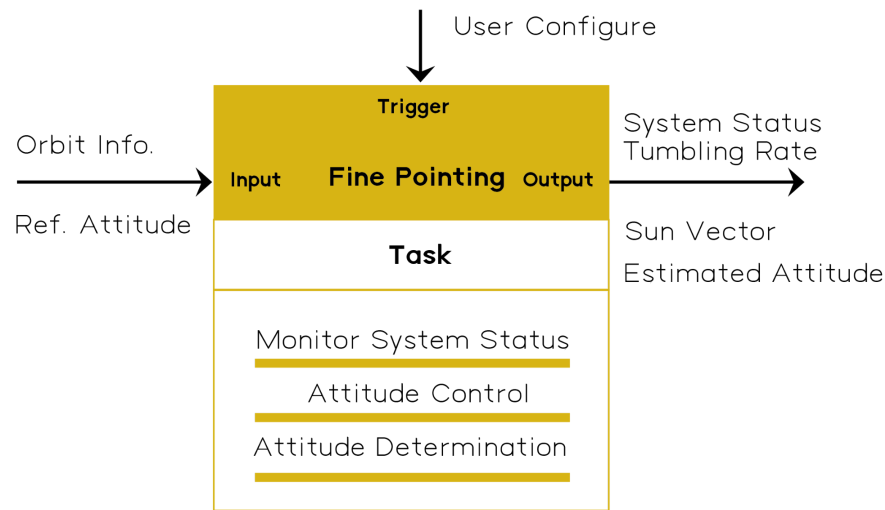


Figure 5-4. Flowchart of Fine Pointing Mode

5.5 LVLH Mode

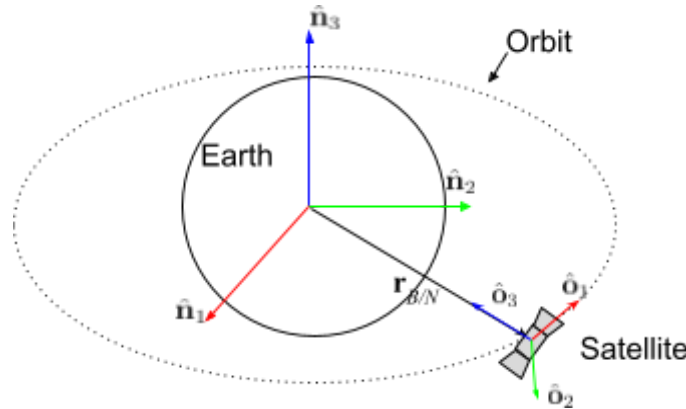


Figure. 5-5 Inertial frame and LVLH frame

Define the Local-Vertical/Local-Horizontal (LVLH) frame as $O = \{\hat{o}_1, \hat{o}_2, \hat{o}_3\}$. To find the basis vectors $\hat{o}_1$, $\hat{o}_2$, and $\hat{o}_3$ represented in N frame, the position of the satellite (B) relative to the earth center (N) in the N frame ${}^N\mathbf{r}_{B/N}$ and the velocity of the satellite relative to the earth center in the N frame ${}^N\mathbf{v}_{B/N}$ should be given. The basis vectors can be derived as

$${}^N\hat{o}_3 = \frac{-{}^N\mathbf{r}_{B/N}}{\|{}^N\mathbf{r}_{B/N}\|} \quad \text{Equation 5-1}$$

$${}^N\hat{o}_2 = -\frac{{}^N\mathbf{r}_{B/N} \times {}^N\mathbf{v}_{B/N}}{\|{}^N\mathbf{r}_{B/N} \times {}^N\mathbf{v}_{B/N}\|} \quad \text{Equation 5-2}$$

$${}^N\hat{o}_1 = \hat{o}_2 \times \hat{o}_3 \quad \text{Equation 5-3}$$

The direction cosine matrix from the reference (R) frame to the N frame is

$${}^N\mathbf{T}_R = [{}^N\hat{o}_1 \quad {}^N\hat{o}_2 \quad {}^N\hat{o}_3] \quad \text{Equation 5-4}$$

The reference attitude $\mathbf{q}_{R/N}$ can be computed from Equation 5-4.

- **Trigger:** The user manually sets the MODE bits in register CONF. When the satellite cannot see the target in Target Tracking Mode, ADCS-x switches to LVLH Mode automatically.
- **Task:** ADCS-x monitors system status, updates register map, performs automatic earth pointing, and aligns the satellite to the orbit velocity vector.
- **Input:** TLE plus current time or GNSS information is required for ADCS-x to determine the LVLH frame.
- **Output:** In every control cycle, the system status e.g. tumbling rate, sun vector, and estimated attitude, etc. are updated from [User\(0\)](#) register map, while sensor and actuator raw data are updated from [Sensor/Actuator\(1\)](#) register map.

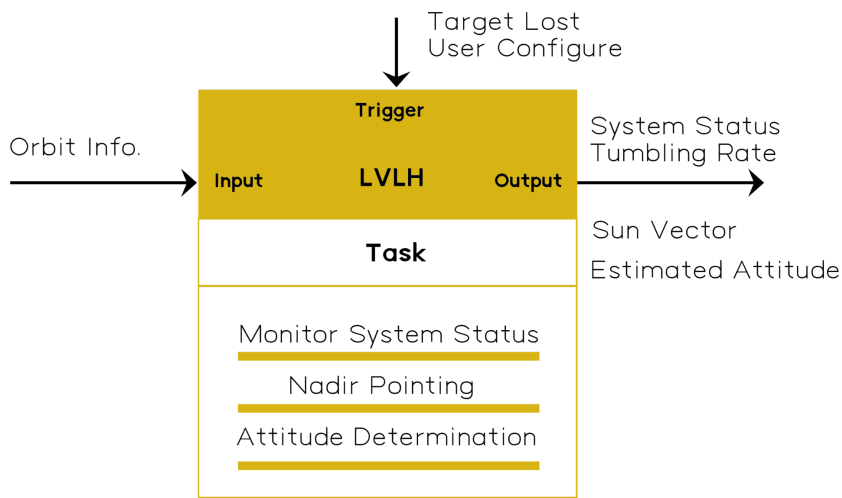


Figure 5-6. Flowchart of LVLH Mode

5.6 Target Tracking Mode

Whenever the mission requires the satellite to stare at a certain location on the earth, the OBC can trigger this Target Tracking Mode. The satellite will be controlled to stare at a target.

To point our satellite accurately at a target on Earth, follow these steps:

1. **Determine Target Vector:** Use Equation 5-5 to calculate the position vector of the target relative to the satellite $\mathbf{r}_{T/B}$.

$${}^N\mathbf{r}_{T/B} = {}^N\mathbf{r}_{T/N} - {}^N\mathbf{r}_{B/N} \quad \text{Equation 5-5}$$

where ${}^N\mathbf{r}_{B/N}$ is the position vector of the satellite relative to the earth center in the ECI frame. ${}^N\mathbf{r}_{T/N}$ is the position vector of the target relative to the earth center in the ECI frame.

2. **Transform to ADCS Frame:** convert ${}^N\mathbf{r}_{T/B}$ to the ADCS frame.

$${}^B\mathbf{r}_{T/B} = {}^B\mathbf{A}_N {}^N\mathbf{r}_{T/B} \quad \text{Equation 5-6}$$

where ${}^B\mathbf{A}_N$ is the transformation matrix from the ECI frame to the ADCS frame.

3. **Normalize the Vector:** Convert the transformed vector into a unit vector.

$${}^B\hat{\mathbf{r}}_{T/B} = \frac{{}^B\mathbf{r}_{T/B}}{\|{}^B\mathbf{r}_{T/B}\|} \quad \text{Equation 5-7}$$

4. **Identify Rotation Axis:** Find the principal rotation axis perpendicular to both the satellite's alignment axis and the unit target vector.

$$\hat{\mathbf{e}} = \frac{{}^B\hat{\mathbf{p}} \times {}^B\hat{\mathbf{r}}_{T/B}}{\|{}^B\hat{\mathbf{p}} \times {}^B\hat{\mathbf{r}}_{T/B}\|} \quad \text{Equation 5-8}$$

5. **Calculate Angular Distance:** Compute the angular distance between the satellite's axis and the target vector.

$$\psi = \cos^{-1}({}^B\hat{\mathbf{p}} \cdot {}^B\hat{\mathbf{r}}_{T/B}) \quad \text{Equation 5-9}$$

6. **Get Attitude Error:** Represent attitude error in quaternion from the angular distance and the rotation axis.

$$\mathbf{q} = \begin{bmatrix} \cos(\psi/2) \\ \hat{e}_1 \sin(\psi/2) \\ \hat{e}_2 \sin(\psi/2) \\ \hat{e}_3 \sin(\psi/2) \end{bmatrix} \quad \text{Equation 5-10}$$

Moreover, if the target exits the boresight of the satellite in the Target Tracking Mode, the ADCS-x will enter the LVLH Mode automatically.

- **Trigger:** The user manually sets the MODE bits in CONF.
- **Task:** ADCS-x monitors system status and updates the register map while

pointing to the target on the earth.

- **Input:** OBC specifies the target's latitude, longitude, and altitude. TLE or GNSS information is required for ADCS-x to determine the reference frame. It is essential to specify a pointing axis ${}^B\hat{\mathbf{p}}$ in the target tracking mode. This command is represented by a unit vector in the ADCS's frame as shown below:

Table 5-2 Pointing Axis Example for Target Tracking Mode

Parameter	POINTING_AXIS_X	POINTING_AXIS_Y	POINTING_AXIS_Z
Value	0	0	1

The user needs to set a pointing axis which is at the index 14~16 in the User(0) register map. The default setting aligns the satellite's Z-axis to the target. However, users can adjust this command to select a different axis for targeting purposes, ensuring flexibility in directing the satellite toward the desired point.

- **Output:** In every control cycle, the system status e.g. tumbling rate, sun vector, and estimated attitude, etc. are updated from [User\(0\)](#) register map, while sensor and actuator raw data are updated from [Sensor/Actuator\(1\)](#) register map.

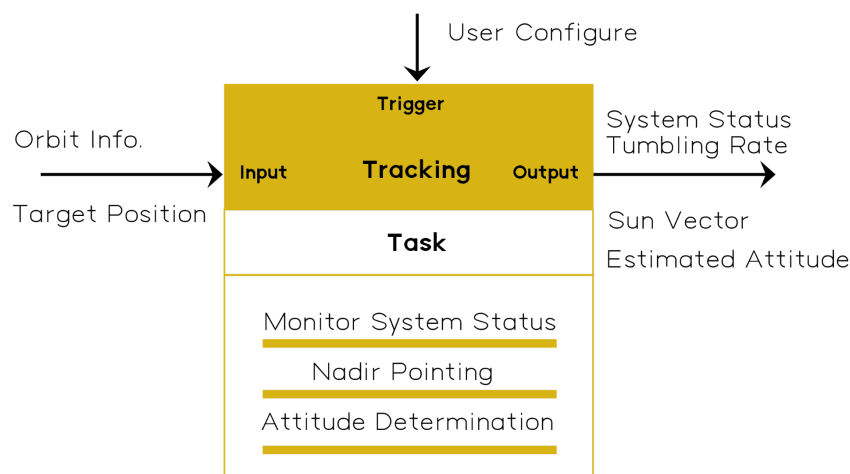


Figure 5-7. Flowchart of Target Tracking Mode

6. SOFTWARE INTERFACE

In this section, the floating-point numbers used in ADCS-x follow the IEEE-754 standard.

6.1 Serial Communication Protocol

The ADCS-x module supports serial communication via UART or RS-485 interfaces, preset to operate at 115200 baud rate with 8N1 configuration. This configuration facilitates half-duplex communication, allowing ADCS-x to function as a subordinate unit, prepared to process master commands for data exchange. The message consists of **Header**, **Body**, and **Checksum**. The format and its composition as shown in Figure 6-1.

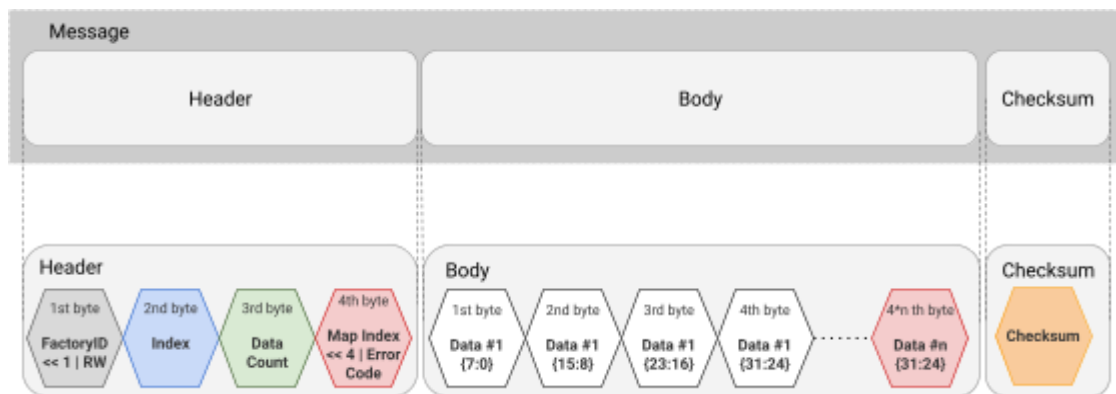


Figure 6-1. Message Format

where the **Header** includes the following components:

1st Byte **Head**:

The first byte initiates a message by left-shifting the **FactoryID** by 1 bit. The LSB of **Head** represents whether it is a read(1) or write(0) message. The **FactoryID** default value is set to 0x64.

2nd Byte **Index**:

The 2nd byte indicates the starting register.

3rd Byte **Data Count**:

The 3rd byte informs the ADCS-x module how many should be read/written.

Remark: It should be noted that this instruction treats 4 bytes as 1 data count.

4th Byte **Map Index and Error Code**:

The 4th byte is composed of 2 parts, Error Code and Map Index.

Table 6-1. The composition of the 4th byte

Map Index	Description	
	Map Index indicates the current register map. The register map is separated into three parts: User (0) , Sensor/Actuator(1) , and Parameter(2) . In this instruction, the value in the map x and index y is represented by (x)-y for short.	
	Map Index	Description
	0b0000	User
	0b0001	Sensor/Actuator
	0b0010	Parameter
Error Code	Description	
	Error code is the response for MRST. In MTSR, the error code should always fill 0b0000, otherwise treated as an invalid header. Map Index indicates the current register map.	
	Error Code	Description
	0b0000	No error, request handled
	0b0001	Checksum error
	0b0010	Invalid register

The **Body** are values which are given from the User command or stored from the register map. The size of the message body is $n*4$ bytes, where n is the value of **Data Count**.

The message ends up with the **Checksum**. The checksum is a byte sum up each byte within the message and then takes 2's complement. The pseudocode should be like the following:

1	<code>generate_checksum(buf, len):</code>	<code>verify_checksum(buf, len):</code>
2	<code>sum = 0</code>	<code>// assume checksum placed in the last byte of buf</code>
3	<code>while(--len):</code>	<code>sum = 0</code>
4	<code>sum+=buf[len]</code>	<code>while(--len):</code>
5	<code>return 0xff - sum+1</code>	<code>sum+=buf[len]</code>
6		<code>return sum == 0</code>

Figure 6-2 and Figure 6-3 are message sequences of register writing and reading. In the writing scene, the user specifies register Index and Map Index in **Header** and

assigns values in **Body**. The ADCS-x verifies the request by checking Map index, register Index and Data Count. The response message from ADCS-x only contains **Header** and **Checksum** in the writing scene. In the reading scene, the user only needs to specify the register index, Map Index and Data Count correctly. The ADCS-x returns message with **Body** if the request is correct.

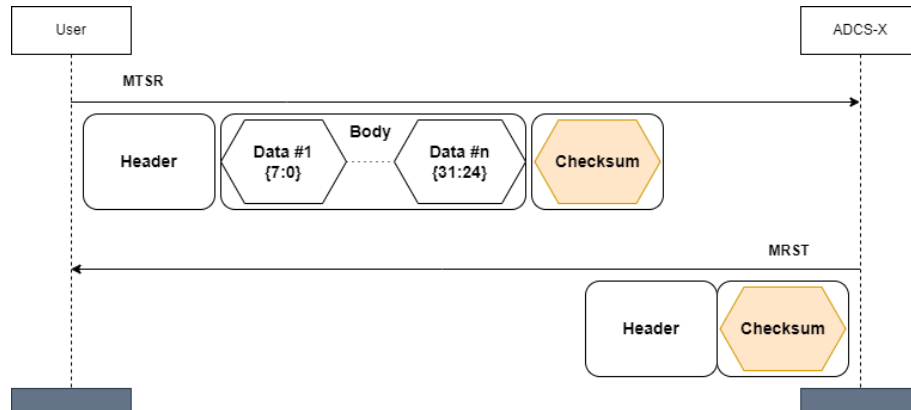


Figure 6-2. Message sequence of register writing

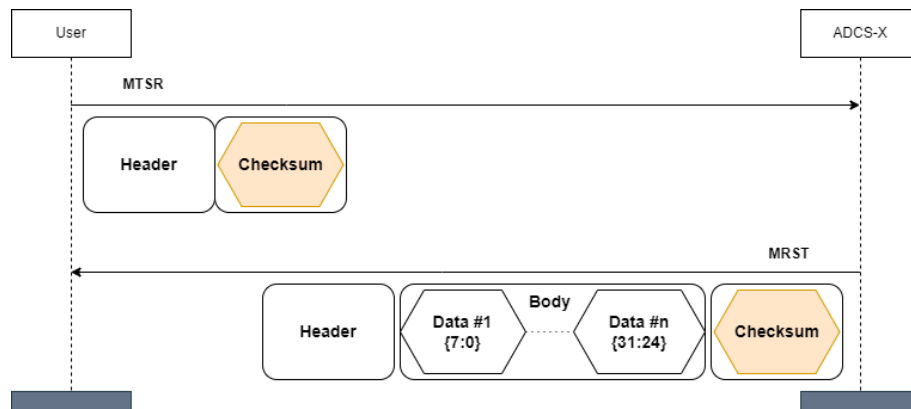


Figure 6-3. Message sequence of register reading

This instruction elaborates the following examples (Examples 6-1 to 6-5) for the User to understand the interaction with ADCS-x.

Example 6-1. Header in the writing scene

The user asks to write MASS into the register, the Header is

1st byte	2nd byte	3rd byte	4th byte
Head	Index	Data Count	Map Error
0xC8	0x00	0x01	0x20

1. The Head is $(0x64 \ll 1) \mid 0x00 = 0xC8$.
2. The MASS index is 0x00, see [Table 6-4](#) for details.
3. The type of MASS is float, which means it needs 1 data count.
4. The MASS is located at the parameter map (map index = 0x02) and the error code should be 0x00 in MTSR. As the result, the 4th byte is $(0x02 \ll 4) \mid 0x00 = 0x20$

Example 6-2. Header in the reading scene

The user asks the SNID result, the Header is

1st byte	2nd byte	3rd byte	4th byte
Head	Index	Data Count	Map Error
0xC9	0x01	0x03	0x00

1. The **Head** is $(0x64 \ll 1) \mid 0x01 = 0xC9$.
2. The SNID index is 0x01, see [Table 6-2](#) for detail.
3. SNID is a char array with 12 bytes, which means it needs 3 data counts.
4. The register SNID which is located at the user map (map index = 0x00) and the error code should be 0x00 in MTSR. As the result, the 4th byte is $(0x00 \ll 4) \mid 0x00 = 0x00$.

Example 6-3. Write MASS value

The user asks to write 15 [kg] into the register MASS. The header is the same as Example 6-1. Because the floating value 15.0 kg can be represented as 0x41700000 in HEX. The message sequence is shown in Figure 6-4.

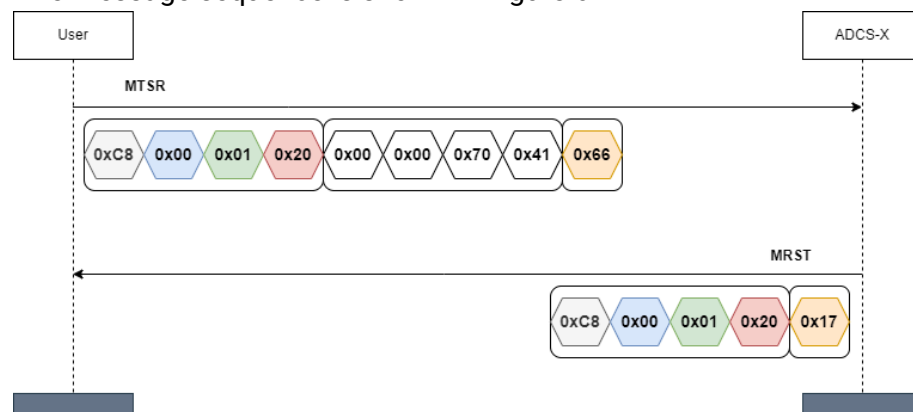


Figure 6-4. writing 15.0 kg into MASS (2)-0

Example 6-4. Read SNID value

An ADCS-x with serial number TAD102001E. The user asks to read this result. Because the serial number has only 10 characters, but the SNID has 12 bytes. The rest of the unused bytes in SNID are replaced 0x00 instead.

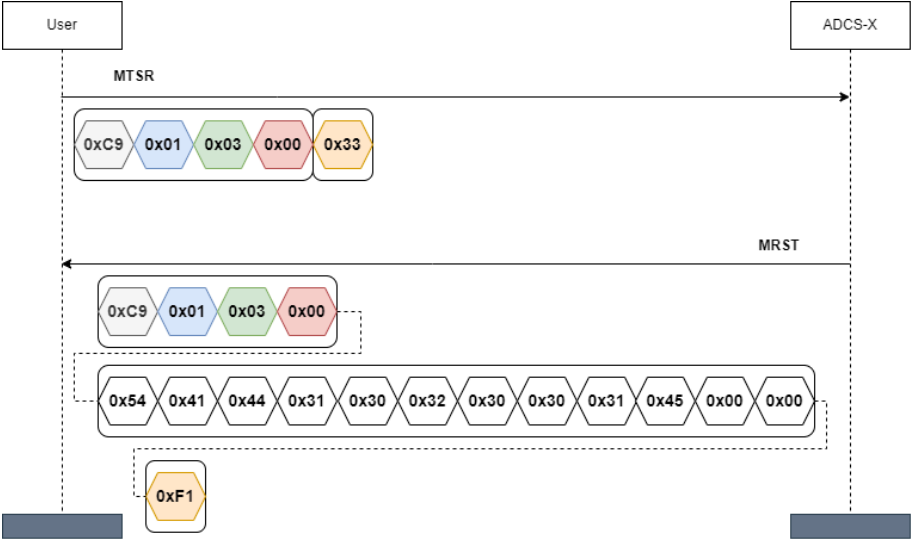


Figure 6-5. Read SNID (0)-1

Example 6-5. Read SNID value with wrong checksum

An ADCS-x with SNID TAD102001E. The user asks to read this result but the requested message has the wrong checksum. The ADCS-x returns error 0x01.

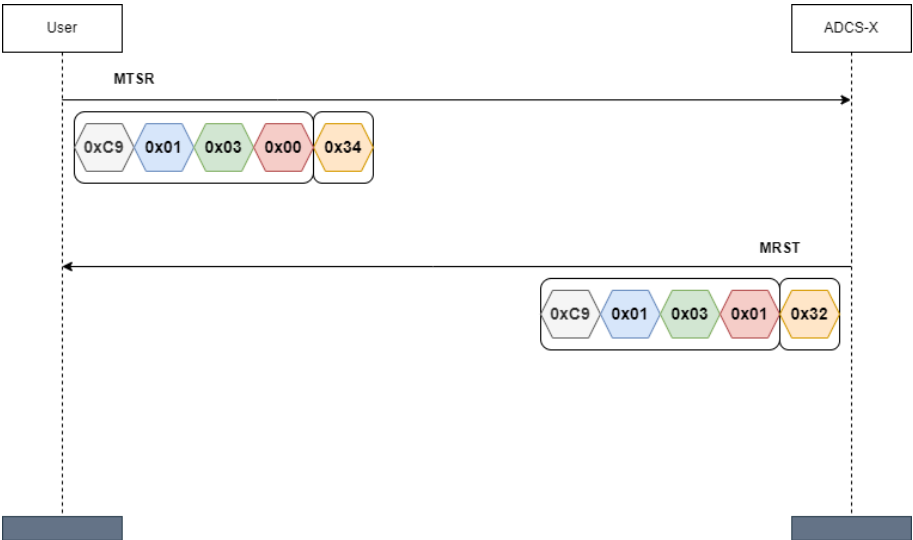


Figure 6-6. Read SNID (0)-1 with wrong checksum

6.2 Register Map

Register map is the application interface where OBC exchanges the data and sets the ADCS-x. The data is stored in the Little-Endian pattern.

Table 6-2. User Register (0)

Index	Alias	Type	Default
0	FACT	uint16 [2]	-
1	SNID	char [12]	-
4	CONF	uint8 [4]	0x00000000
5	TIME	uint8 [4]	0x00000000
6	DATE	uint8 [4]	0x00000000
7	LLA_REF	float [3]	[0, 0, 0]
10	Q_REF	float [4]	[1, 0, 0, 0]
14	POINTING_AXIS	float [3]	[0, 0, 1]
17	TLE	char [140]	-
52	GGA	char [128]	-
84	ZDA	char [64]	-
100	SV_USER	float [3]	[0, 0, 0]
103	MTQ_USER	float [3]	[0, 0, 0]
106	CMG0_G_RATE_USER	float	0
107	CMG0_W_TOR_USER	float	0
108	CMG1_G_RATE_USER	float	0
109	CMG1_W_TOR_USER	float	0
110	CMG2_G_RATE_USER	float	0
111	CMG2_W_TOR_USER	float	0
112	CMG3_G_RATE_USER	float	0
113	CMG3_W_TOR_USER	float	0
128	STAT	uint8 [4]	0x00000000
129	ACT_ERR	uint8 [4]	0x00000000
130	SEN_ERR	uint8 [4]	0x00000000
131	CSS_ERR	uint8 [4]	0x00000000
132	Q	float [4]	[1, 0, 0, 0]
136	RATE	float [3]	[0, 0, 0]
139	LLA	float [3]	[0, 0, 0]
142	ATT_ERROR	float [3]	[0, 0, 0]
145	ATT_ERROR_RATE	float [3]	[0, 0, 0]

Index	Alias	Type	Default
148	ADCS_TMP	int16 [2]	0x00000000
149	CMG0_TMP	int16 [2]	0x00000000
150	CMG1_TMP	int16 [2]	0x00000000
151	CMG2_TMP	int16 [2]	0x00000000
152	CMG3_TMP	int16 [2]	0x00000000
153	FSS_TMP1	int16 [2]	0x00000000
154	FSS_TMP2	int16 [2]	0x00000000
155	FSS_TMP3	int16 [2]	0x00000000
156	SV	float [3]	[0, 0, 0]
159	MAG	float [3]	[0, 0, 0]

Legend:

W = Writable

R = Readable

U = Unimplemented, read as 0

FACT

Bit 31- Bit 16	Bit 15 - Bit 0
-	VER
-	R

VER: Firmware Version

SNID

Byte 0 - Byte 11
SNID
R

SNID: Serial number

CONF

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
TARGET_ELEV							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
-	-	-	-	-	-	-	-
U							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
-	-	-	-	-	-	-	-
U							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	-	-	-	-	MODE		
U					RW		

TARGET_ELEV: Target elevation angle [degree]

MODE: Mode bits

8 = BBQ mode(TBD)

111 = manual mode

110 = Sun spin

101 = Target Tracking Mode

100 = LVLH Mode

011 = Fine Pointing Mode

010 = Sun-Pointing Mode

001 = De-tumbling Mode

000 = Safe Mode

TIME

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
HR10				HR01			
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
MIN10				MIN01			
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
SEC10				SEC01			
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-							
U							

HR10: Binary-Coded Decimal Value of Hours bits, 10 digits; contain a value from 0 to 2

HR01: Binary-Coded Decimal Value of Hours bits, 1 digit; contain a value from 0 to 9

MIN10: Binary-Coded Decimal Value of Minutes bits, 10 digits; contain a value from 0 to 5

MIN01: Binary-Coded Decimal Value of Minutes bits, 1 digits; contain a value from 0 to 9

SEC10: Binary-Coded Decimal Value of Seconds bits, 10 digits; contain a value from 0 to 5

SEC01: Binary-Coded Decimal Value of Seconds bits, 1 digits; contain a value from 0 to 9

DATE

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
YEAR10				YEAR01			
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
MONTH10				MONTH01			
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
DAY10				DAY01			
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-				WDAY01			
U				RW			

YEAR10: Binary-Coded Decimal Value of Years bits, 10 digits

YEAR01: Binary-Coded Decimal Value of Years bits, 1 digit

MONTH10: Binary-Coded Decimal Value of Months bits, 10 digits; contain a value from 0 to 1

MONTH01: Binary-Coded Decimal Value of Months bits, 1 digits; contain a value from 0 to 9

DAY10: Binary-Coded Decimal Value of Days bits, 10 digits; contain a value from 0 to 3

DAY01: Binary-Coded Decimal Value of Days bits, 1 digits; contain a value from 0 to 9

WDAY01: Binary-Coded Decimal Value of Weekdays bits, 1 digits; contain a value from 0 to 6

LLA_REF

float [0]	float [1]	float [2]
LATR	LONGR	ALTR
RW	RW	RW

LATR/LONGR/ALTR: Reference Latitude/Longitude/Altitude input for Target Tracking mode
Before entering the Target Tracking Mode, OBC should write the target's position in LATR/LONGR/ALTR registers. The unit of LATR and LONGR is degree, and the unit of ALTR is km.

Q_REF

float [0]	float [1]	float [2]	float [3]
Q0REF	Q1REF	Q2REF	Q3REF
RW	RW	RW	RW

Q0REF/Q1REF/Q2REF/Q3REF: Reference Quaternion

In Fine Pointing Mode, ADCS-x sets **RFRM** bits in **CONF** and **Q0REF/Q1REF/Q2REF/Q3REF** as the reference for attitude control.

POINTING_AXIS

float [0]	float [1]	float [2]
POINTING_AXIS_X	POINTING_AXIS_Y	POINTING_AXIS_Z
RW	RW	RW

POINTING_AXIS_X/Y/Z: The pointing axis of the ADCS frame for the target/sun tracking mode.

TLE

char [0] - char [68]	char [69] - char [137]	char [138] - char [139]
LINE_1	LINE_2	-
RW	RW	RW

TLE: Character buffer for NORAD standard Two-Line Orbital Element Set Format input, max 35 word (140 bytes)

Example:

```
"1 25544U 98067A 08264.51782528 -.00002182 00000-0 -11606-4 0 2927
2 25544 51.6416 247.4627 0006703 130.5360 325.0288 15.72125391563537"
```

GGA

char [0] - char [127]
GP GGA
RW

GP GGA: Character buffer for GNSS GP GGA input, max. 32 words (128 bytes).

Example:

```
"$GP GGA,hhmmss.ssssss,ddmm.mmmmm,a,dddmm.mmmmm,a,x,xx,x.x,x.x,M,x.x,M,x.x,xxx
x*hh<CR><LF>"
```

Note that when ADCS-x is equipped with GNSS module (i.e. **STAT.GNSS = 0b11**), ADCS-x computer always uses its own GNSS data, and will automatically omit GGA or ZDA write operations from the OBC. However, once there is a GNSS hard error (i.e. **STAT.GNSS = 0b10**), ADCS-x requires the OBC to provide the GNSS data and continue to operate in advanced mode.

ZDA

char [0] - char [63]
ZDA Word Buffer
RW

GNZDA: Character buffer for GNSS ZDA input, max. 16 words (64 bytes).

Example "\$GNZDA,hh mm ss.ssssss,xx,xx,xxxx,xx,xx*hh<CR><LF>"

Similar to the GGA register, the ADCS-x computer may ignore OBS ZDA write operation if ADCS-x has its own GNSS module equipped. Further details see the description of the register GGA.

SV_USER

float [0]	float [1]	float [2]
SV_USER_X	SV_USER_Y	SV_USER_Z
RW	RW	RW

SV_USER_X/Y/Z: Sun pointing vector in the body frame, as specified by the user if such a request is made. The vector is a unit vector.

MTQ_USER

float [0]	float [1]	float [2]
MTQ_0_USER	MTQ_1_USER	MTQ_2_USER
R/W	R/W	R/W

MTQ_x_USER: custom magnetorquer dipole [A-m²], x=0,1,2

CMGx_G_RATE_USER

float
CMGx_G_RATE_USER
R

CMGx_G_RATE_USER: The custom gimbal rate of the xth CMG, where x=0,1,2,3. Unit is rad/s.

CMGx_W_TOR_USER

float
CMGx_W_TOR_USER
R

CMGx_W_TOR_USER: The custom wheel torque of the xth CMG, where x=0,1,2,3. The unit is mNm.

STAT

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
HERR	SERR	WDT	UV	OC	OT	GNSS	
R	R	R	R	R	R	R	R
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
-	-	-	-	-	-	-	-
U							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
TLE	DES	SUN	TGL	TUMB	AME	CUSSV	EKF
R	R	R	R	R	R	U	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
MODE			-	-	-	-	-
R			U				

HERR:

1 = Hard error detected.

0 = No hard error detected.

SERR:

1 = Soft error detected.

0 = No software error detected.

WDT: Watchdog Timer Timeout Flag

1 = Watchdog timer timeout, ADCS-x reset to safe mode

0 = No watchdog timeout detected

UV: Undervoltage Flag

1 = Undervoltage detected

0 = No voltage error

OC: Overcurrent Protection Flag

1 = Overcurrent detected, ADCS-x switches back to Safe Mode

0 = No overcurrent detected

OT: Over temperature or Under Temperature Protection Flag

1 = Over temperature or under temperature detected, ADCS-x switches back to Safe Mode

0 = No over or under temperature detected

GNSS: GNSS Status bits

Bit 25: 1 = GNSS over current, 0: Normal

Bit 24: 1 = up-to-date, 0 = out-of-date

TLE: TLE update status

0: out-of-date

1: up-to-date

DES: De-saturation Flag

When the inner rotor speed of the rotor of the CMG reaches its saturation speed, the ADCS-x begins its de-saturation operation. In this period, this flag is raised, and the attitude control accuracy usually degrades.

1 = The angular momentum stored in the CMG hits its limit, or the ADCS-x is de-saturating the CMG while executing the attitude control task for the user.

0 = Angular momentum below the threshold, and the ADCS-x is not de-saturating while controlling the satellite's attitude.

SUN: Sun Detected Flag

1 = Sun detected, sun vector is valid

0 = Sun not detected. The sun vector is invalid. It usually means the satellite is entering an eclipse area of the orbit

TGL: Target Lost Flag

1 = Target lost, mandatorily switch to LVLH Mode

0 = Tracking target or not in Target Tracking Mode

TUMB: Tumbling Flag

1 = Body angular velocity of the satellite is detected to be higher than 5 deg/second

0 = No tumbling present

AME: Angular Momentum Exceeded Flag

1 = Angular momentum exceeded in Safe Mode, mandatorily switch to De-tumbling Mode

0 = Angular momentum below threshold or not in Safe Mode

CUSSV: Custom Sun Vector Supported Flag

1 = Custom sun vector is supported

0 = Custom sun vector is not supported

EKF: EKF flag

1: active, 0: non-active

MODE: Current Running Mode bits

111 = Reserved

110 = Reserved

101 = Target Tracking Mode

100 = LVLH Mode

011 = Fine Pointing Mode

010 = Sun-Pointing Mode

001 = De-tumbling Mode

000 = Safe Mode

ACT_ERR

Bit 31 - Bit 24				
-				
U				
Bit 23 - Bit 16				
-				
U				
Bit 15 - Bit 11		Bit 10	Bit 9	Bit 8
-		MTQ2	MTQ1	MTQ0
U		R	R	R
Bit 7 - Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	CMG3	CMG2	CMG1	CMG0
U	R	R	R	R

MTQ0/1/2: Error flag of the magnetorquer.

1 = Magnetorquer hard error detected. Possibly due to

- Magnetorquer over-current
- Magnetorquer under-current

0 = No hard error detected

CMG0/1/2/3: Error flag of the CMG

1 = CMG hard error detected

0 = No hard error detected

SEN_ERR

Bit 31 - Bit26				Bit 25	Bit 24	
-				STR1	STR0	
U				R	R	
Bit 23 - Bit 20			Bit 19	Bit 18	Bit 17	Bit 16
-			IMU3	IMU2	IMU1	IMU0
U			R	R	R	R
Bit 15 - Bit14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
-	MAG5	MAG4	MAG3	MAG2	MAG1	MAG0
U	R	R	R	R	R	R
Bit 7 - Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	FSS5	FSS4	FSS3	FSS2	FSS1	FSS0
U	R	R	R	R	R	R

STRx: Error flag of the star tracker, x=0,1.

1: STR error detected

0: No error

IMUx: IMU Hard Error Flag, x = 0,1,2,3

1 = IMU hard error detected

0 = No hard error detected

MAGx: Magnetometer Error Flag, x=0,1,2,3,4,5.

1 = Magnetometer error detected

0 = No error present

FSSx: Fine Sun Sensor Error Flag, x=0,1,2,3,4,5.

1 = Fine sun sensor error detected

0 = No error present

CSS_ERR

Bit 31 - Bit 24							
U							
Bit 23 - Bit 16							
U							
Bit 15 - Bit11					Bit 10	Bit 9	Bit 8
U					CSS_10	CSS_9	CSS_8
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CSS_7	CSS_6	CSS_5	CSS_4	CSS_3	CSS_2	CSS_1	CSS_0

CSSx: Error flag of CSS, x=0,1,...10.

1: error detected

0: no error detected

Q

float [0]	float [1]	float [2]	float [3]
Q0	Q1	Q2	Q3
R	R	R	R

Q0/Q1/Q2/Q3: Attitude Quaternion

According to **RFRM** bits, ADCS-x estimates the current attitude quaternion in the corresponding frame, then stored in **Q0/Q1/Q2/Q3** registers. Please remember that unsynchronized **TIME**, **DATE**, obsolete TLE, or GNSS (if applicable) makes the result of attitude determination invalid.

RATE

float [0]	float [1]	float [2]
RATEX	RATEY	RATEZ
R	R	R

RATEX/RATEY/RATEZ: Attitude rate in body frame [rad/s].

LLA

float [0]	float [1]	float [2]
LAT	LONG	ALT
R	R	R

LAT/LONG/ALT: Current position of the satellite in geodetic coordinate, where LAT is latitude, LONG is longitude and ALT is altitude. When the GNSS module is equipped, **LAT/LONG/ALT** registers to store the current position acquired from the GNSS module. Otherwise, **LAT/LONG/ALT** are propagated from TLE, **TIME**, and **DATE**.

The unit of **LAT/LONG** is degree and the unit of **ALT** is km.

ATT_ERROR

float [0]	float [1]	float [2]
ATT_ERROR_X	ATT_ERROR_Y	ATT_ERROR_Z
R	R	R

ATT_ERROR_X/Y/Z: attitude error of the body frame relative to inertial frame, represented in MRP.

ATT_ERROR_RATE

float [0]	float [1]	float [2]
ATT_ERROR_RATE_X	ATT_ERROR_RATE_Y	ATT_ERROR_RATE_Z
R	R	R

ATT_ERROR_RATE_X/Y/Z: attitude error rate of the body frame relative to inertial frame in body frame. The unit is rad/s.

ADCS_TMP

Bit 31 - Bit 16	Bit 15 - Bit 0
-	BRDTMP
-	R

BRDTMP: ADCS-x Main Board Temperature, stored in 16 bits signed integer.

$$\text{Temperature } [^{\circ}\text{C}] = \text{BRDTMP} \times 150 / 32768 \quad \text{Equation 6-1}$$

If ADCS-x fails to communicate with the temperature sensor, **BRDTMP** reads as 0xFFFF.

CMGx_TMP

Bit 31 - Bit 16	Bit 15 - Bit 0
TMP2	TMP1
R	R

The xth CMG temperatures, x=0,1,2,3. Each CMG has two thermometers.

TMP1: 1st thermocouple, stored in 16 bits signed integer.

TMP2: 2nd thermocouple, stored in 16 bits signed integer.

$$\text{Temperature } [^{\circ}\text{C}] = \text{CMGTMP} \times 150 / 32768 \quad \text{Equation 6-2}$$

If ADCS-x fails to communicate with the temperature sensor, **CMGTMP** reads as 0xFFFF.

FSS_TMP1

Bit 31 - Bit 16	Bit 15 - Bit 0
FSS1TMP	FSS0TMP
R	R

FSS_TMP2

Bit 31 - Bit 16	Bit 15 - Bit 0
FSS3TMP	FSS2TMP
R	R

FSS_TMP3

Bit 31 - Bit 16	Bit 15 - Bit 0
FSS5TMP	FSS4TMP
R	R

FSS0TMP/ FSS1TMP/ FSS2TMP/ FSS3TMP/ FSS4TMP/ FSS5TMP: Fine sun sensor temperature, stored in 16 bits signed integer. How FSS0TMP, FSS1TMP, FSS2TMP, FSS3TMP, FSS4TMP, and FSS5TMP map to each fine sun sensor should be set by the user in support software.

$$\text{Temperature } [^{\circ}\text{C}] = \text{FSSxTMP} \times 0.03125 \quad \text{Equation 6-3}$$

SV

float [0]	float [1]	float [2]
SV_X	SV_Y	SV_Z
R	R	R

SV_X/Y/Z: Sun pointing vector in body frame. The vector is a unit vector.

Stores the current sun vector in the body frame. If the sun isn't detected, SVX/SVY/SVZ reads as a null vector. Otherwise, SVX/SVY/SVZ comprise a unit vector.

MAG

float [0]	float [1]	float [2]
MAG_X	MAG_Y	MAG_Z
R	R	R

MAG_X/Y/Z: A Magnetic field vector in body frame, unit: nT.

Table 6-3. Sensor/Actuator Register (1)

Index	Alias	Type	Default
0	MAG_MAT	float [9]	-
9	MAG_VEC	float [3]	-
12	MAG_STAT	uint8 [4]	0x00000000
13	MAG0_S	float [3]	0
16	MAG1_S	float [3]	0
19	MAG2_S	float [3]	0
22	MAG3_S	float [3]	0
25	MAG4_S	float [3]	0
28	MAG5_S	float [3]	0
31	FSS_STAT	uint8 [4]	0x00000000
32	FSS0_SV	uint16 [2]	0x00000000
33	FSS0_PDSUM	uint16 [2]	0x00000000
34	FSS1_SV	uint16 [2]	0x00000000
35	FSS1_PDSUM	uint16 [2]	0x00000000
36	FSS2_SV	uint16 [2]	0x00000000
37	FSS2_PDSUM	uint16 [2]	0x00000000
38	FSS3_SV	uint16 [2]	0x00000000
39	FSS3_PDSUM	uint16 [2]	0x00000000
40	FSS4_SV	uint16 [2]	0x00000000
41	FSS4_PDSUM	uint16 [2]	0x00000000
42	FSS5_SV	uint16 [2]	0x00000000
43	FSS5_PDSUM	uint16 [2]	0x00000000

44	IMU_STAT	uint8 [4]	0x00000000
45	IMU0_S	float [3]	0
48	IMU1_S	float [3]	0
51	IMU2_S	float [3]	0
54	IMU3_S	float [3]	0
57	STR_STAT	uint8 [4]	0x00000000
58	STR0_S	float [4]	0
62	STR1_S	float [4]	0
66	CSS	float [12]	0
78	CMG_STAT	uint8 [4]	0x00000000
79	CMG0_G_ANGLE	float	0
80	CMG0_W_RATE	float	0
81	CMG1_G_ANGLE	float	0
82	CMG1_W_RATE	float	0
83	CMG2_G_ANGLE	float	0
84	CMG2_W_RATE	float	0
85	CMG3_G_ANGLE	float	0
86	CMG3_W_RATE	float	0
87	MTQ	float [3]	0
90	CMG0_G_RATE	float	0
91	CMG0_W_ACC	float	0
92	CMG1_G_RATE	float	0
93	CMG1_W_ACC	float	0
94	CMG2_G_RATE	float	0
95	CMG2_W_ACC	float	0
96	CMG3_G_RATE	float	0
97	CMG3_W_ACC	float	0

MAG_MAT

float [0] ~ float [8]
MAG_MAT_x
R

MAG_MAT_0~8: 3 x 3 Magnetic Sensor calibration matrix **C**, with elements C_{00} , C_{01} , C_{02} , C_{10} , C_{11} , C_{12} , C_{20} , C_{21} , C_{22} , represented as **MAG_MAT_x**, in 32-bit float, x=0,1,2,3,4,5,6,7,8.

MAG_VEC

float [0]	float [1]	float [2]
MAG_VEC_0	MAG_VEC_1	MAG_VEC_2
R	R	R

MAG_VEC_0/1/2: 3 x 1 Magnetic Sensor offset calibration vector, in 32-bit float, x=0,1,2.

MAG_STAT

Bit 31-Bit 24						
U						
Bit 23-Bit 16						
U						
Bit 15-Bit 8						
U						
Bit 7-Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	MAG_5_INS TALL	MAG_4_INS TALL	MAG_3_INS TALL	MAG_2_INS TALL	MAG_1_INS TALL	MAG_0_INST ALL
U	R	R	R	R	R	R

MAG_x_INSTALL: the install status of the MAG_x, x=0,1,2,3,4,5.

0: uninstalled; 1: installed.

MAGx_S

float [0]	float [1]	float [2]
MAGx_S_0	MAGx_S_1	MAGx_S_2
R	R	R

MAGx_S_0/1/2: 3 x 1 Magnetic field on three axes in the sensor frame, represented in 32-bit float, x=0,1,2,3,4,5. Unit is Tesla.

FSS_STAT

Bit 31-Bit 24						
U						
Bit 23-Bit 16						
U						
Bit 15 - Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
-	FSS_5_ACTIVE	FSS_4_ACTIVE	FSS_3_ACTIVE	FSS_2_ACTIVE	FSS_1_ACTIVE	FSS_0_ACTIVE
U	R	R	R	R	R	R
Bit 7-Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	FSS_5_INSTALL	FSS_4_INSTALL	FSS_3_INSTALL	FSS_2_INSTALL	FSS_1_INSTALL	FSS_0_INSTALL
U	R	R	R	R	R	R

FSS_x_ACTIVE: the active status of the FSS_x, x=0,1,2,3,4,5.

FSS_x_INSTALL: the install status of the FSS_x, x=0,1,2,3,4,5.

0: uninstalled; 1: installed.

FSSx_SV

Bit 31 - Bit 16	Bit 15 - Bit 0
THETA	PHI
R	R

THETA: Polar angle of incident light of the FSSx, x = 0, 1, 2, 3, 4, 5.

$$\theta = 0.5\pi / 65535 \times \text{THETA} \quad \text{Equation 6-4}$$

PHI: Azimuthal angle of incident light of the FSSx, x=0,1,2,3,4,5.

$$\phi = 2\pi / 65535 \times \text{PHI} \quad \text{Equation 6-5}$$

FSSx_PDSUM

Bit 31 - Bit 16	Bit 15 - Bit 0
-	PDSUM
U	R

FSSx_PDSUM: Total current sensed from FSSx, x=0,1,2,3,4,5.

$$(i_a + i_b + i_c + i_d) = \text{PDSUM} / 3276.8 \quad \text{Equation 6-6}$$

IMU_STAT

Bit 31 - Bit 24				
U				
Bit 23 - Bit 16				
U				
Bit 15 - Bit 8				
U				
Bit 7 - Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	IMU_3_INSTALL	IMU_2_INSTALL	IMU_1_INSTALL	IMU_0_INSTALL
U	R	R	R	R

IMU_x_INSTALL: the install status of the IMU_x, x=0,1,2,3.

0: uninstalled; 1: installed.

IMUx_S

float [0]	float [1]	float [2]
IMUx_S_0	IMUx_S_1	IMUx_S_2
R	R	R

IMUx_S_0/1/2: angular velocity measurement of 3 axes in the sensor frame, x=0,1,2,3. The unit is rad/s.

STR_STAT

Bit 31-Bit 24		
U		
Bit 23-Bit 16		
U		
Bit 15-Bit 8		
U		
Bit 7-Bit 2	Bit 1	Bit 0
U	STR_1_INSTALL	STR_0_INSTALL

STR_x_INSTALL: the install status of the STR_x, x=0,1.

0: uninstalled; 1: installed.

STRx_S

float [0]	float [1]	float [2]	float [3]
STRx_S_0	STRx_S_1	STRx_S_2	STRx_S_3
R	R	R	R

STRx_S_0/1/2/3: quaternion measurement from star tracker, x=0,1.

CSS

float[0] - float[11]
CSS0 - CSS11
R

CSSx: sun intensity from coarse sun sensor, x=0,1,2...11.

CMG_STAT

Bit 31-Bit 24				
U				
Bit 23-Bit 16				
U				
Bit 15 - Bit 8				
U				
Bit 7-Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
-	CMG_3_INSTALL	CMG_2_INSTALL	CMG_1_INSTALL	CMG_0_INSTALL
U	R	R	R	R

CMG_x_INSTALL: the install status of the CMG_x, x=0,1,2,3.

0: uninstalled; 1: installed.

CMGx_G_ANGLE

float
CMGx_G_ANGLE
R

CMGx_G_ANGLE: The gimbal position on the stator frame, x=0,1,2,3.

$$\gamma = \text{CMGx_G_ANGLE} / 65536 \times 2\pi$$

Equation 6-7

CMGx_W_RATE

float
CMGx_W_RATE
R

CMGx_W_RATE: CMG wheel speed, x=0,1,2,3.

$$\Omega = \text{CMGx_W_RATE} / 65536 \times 400\pi \quad \text{Equation 6-8}$$

where the unit of Ω is rad/s and γ is radian.

MTQ

float [0]	float [1]	float [2]
MTQ_0	MTQ_1	MTQ_2
R	R	R

MTQ_x: magnetorquer dipole, x=0,1,2.

CMGx_G_RATE

float
CMGx_G_RATE
R

CMGx_G_RATE: The gimbal rate of the rotor, x=0,1,2,3. The unit is rad/s

CMGx_W_ACC

float
CMGx_W_ACC
R

CMGx_W_ACC: The wheel acceleration, x=0,1,2,3. The unit is rad/s²

Table 6-4. Parameter Register (2)⁴

Index	Alias	Type	Default
0	MASS	float	-
1	INE_TEN	float [9]	0
10	POS_HB_B	float [3]	[0,0,0]
13	ORIEN_HB	float [4]	[1,0,0,0]
17	MAG_INFO	uint8 [4]	-
18	MAG0_ORIEN_BS	float [4]	[1,0,0,0]
22	MAG1_ORIEN_BS	float [4]	[1,0,0,0]
26	MAG2_ORIEN_BS	float [4]	[1,0,0,0]
30	MAG3_ORIEN_BS	float [4]	[1,0,0,0]
34	MAG4_ORIEN_BS	float [4]	[1,0,0,0]
38	MAG5_ORIEN_BS	float [4]	[1,0,0,0]
42	FSS_INFO	uint8 [4]	-
43	FSS0_ORIEN_BS	float [4]	[1,0,0,0]
47	FSS1_ORIEN_BS	float [4]	[1,0,0,0]
51	FSS2_ORIEN_BS	float [4]	[1,0,0,0]
55	FSS3_ORIEN_BS	float [4]	[1,0,0,0]
59	FSS4_ORIEN_BS	float [4]	[1,0,0,0]
63	FSS5_ORIEN_BS	float [4]	[1,0,0,0]
67	IMU_INFO	uint8 [4]	-
68	IMU0_ORIEN_BS	float [4]	[1,0,0,0]
72	IMU1_ORIEN_BS	float [4]	[1,0,0,0]
76	IMU2_ORIEN_BS	float [4]	[1,0,0,0]
80	IMU3_ORIEN_BS	float [4]	[1,0,0,0]
84	STR_INFO	uint8 [4]	-
85	STR0_ORIEN_BS	float [4]	[1,0,0,0]
89	STR1_ORIEN_BS	float [4]	[1,0,0,0]
255	NVM	uint8 [4]	0x00000000

⁴ The register value is initially set to the in-house default value by Tensor Tech in advance, and it can be modified based on user requirements before the mission launches. However, once the satellite is launched into orbit, the register value cannot be reconfigured.

MASS

float
MASS
RW

MASS: The total mass with unit kg.

INE_TEN

float[0] - float[8]
INE_TEN_0 ~8
RW

INE_TEN_0 ~8: The inertia tensor $I_{00}, I_{01}, I_{02}, I_{10}, I_{11}, I_{12}, I_{20}, I_{21}, I_{22}$ of the satellite with unit kg-m².

POS_HB_B

float[0]	float[1]	float[2]
POS_HB_B_0	POS_HB_B_1	POS_HB_B_2
RW	RW	RW

POS_HB_B_0/1/2: The position of the hub relative to the ADCS (B) frame in the ADCS (B) frame.

ORIEN_HB

float[0]	float[1]	float[2]	float[3]
ORIEN_HB_0	ORIEN_HB_1	ORIEN_HB_2	ORIEN_HB_3
RW	RW	RW	RW

ORIEN_HB_0/1/2/3: The orientation of the hub frame relative to the ADCS (B) frame, represented in quaternion.

MAG_INFO

Bit 31-Bit 24
AMOUNT
R/W
Bit 23-Bit 16
U
Bit 15-Bit 8
U
Bit 7-Bit 0
U

AMOUNT: the amount of the installed magnetometer, max. is 6.

MAGx_ORIEN_BS

float[0]	float[1]	float[2]	float[3]
MAGx_ORIEN_BS_0	MAGx_ORIEN_BS_1	MAGx_ORIEN_BS_2	MAGx_ORIEN_BS_3
RW	RW	RW	RW

MAGx_ORIEN_BS_0/1/2/3: The orientation of the ADCS frame relative to the magnetometer (S) frame, represented in quaternion. Where x=0,1,2,3,4,5.

FSS_INFO

Bit 31-Bit 24
AMOUNT
R/W
Bit 23-Bit 16
FOV
R/W
Bit 15 - Bit 8
U
Bit 7-Bit 0
U

AMOUNT: the amount of the installed fine sun sensor, max. is 6.

FOV: the field of view of the FSS [degree].

FSSx_ORIEN_BS

float [0]	float [1]	float [2]	float [3]
FSSx_ORIEN_BS_0	FSSx_ORIEN_BS_1	FSSx_ORIEN_BS_2	FSSx_ORIEN_BS_3
RW	RW	RW	RW

FSSx_ORIEN_BS_0/1/2/3: The orientation of the ADCS frame relative to the FSS (S) frame, represented in quaternion. Where x=0,1,2,3,4,5.

IMU_INFO

Bit 31 - Bit 24
AMOUNT
R/W
Bit 23 - Bit 16
U
Bit 15 - Bit 8
U
Bit 7 - Bit 0
U

AMOUNT: the amount of the installed IMU, max. is 4.

IMUx_ORIEN_BS

float [0]	float [1]	float [2]	float [3]
IMUx_ORIEN_BS_0	IMUx_ORIEN_BS_1	IMUx_ORIEN_BS_2	IMUx_ORIEN_BS_3
RW	RW	RW	RW

IMUx_ORIEN_BS_0/1/2/3: The orientation of the ADCS frame relative to the IMU (S) frame, represented in quaternion. Where x=0,1,2,3.

STR_INFO

Bit 31-Bit 24
AMOUNT
Bit 23-Bit 16
U
Bit 15-Bit 8
U
Bit 7-Bit 0
U

AMOUNT: the amount of the installed star tracker, max. is 2.

STRx_ORIEN_BS

float [0]	float [1]	float [2]	float [3]
STRx_ORIEN_BS_0	STRx_ORIEN_BS_1	STRx_ORIEN_BS_2	STRx_ORIEN_BS_3
RW	RW	RW	RW

STRx_ORIEN_BS_0/1/2/3: The orientation of the ADCS frame relative to the Star Tracker (S) frame, represented in quaternion. Where x=0,1.

NVM

Bit 31 - Bit 24		
U		
Bit 23 - Bit 16		
U		
Bit 15 - Bit 8		
U		
Bit 7 - Bit 1		Bit 0
-	STATUS	S_REQ
U	R	RW

S_REQ: NVM update request

1 = request to store [Parameter Map\(2\)](#) into NVM. Upon completion of the NVM update process, this bit is cleared, triggering an immediate ADCS reset.

0 = no action

STATUS: NVM status

1 = NVM is currently updating. The user is advised to await the clearing of this bit before proceeding.

0 = No Action or NVM update is complete.

7. APPLICATION NOTES: STANDARD OPERATING PROCEDURES

IMPORTANT: Upon unpacking the ADCS-x, it is imperative to utilize Tensor Tech's customer support software for the proper configuration of the ADCS-x. Failure to follow this procedure may result in the ADCS-x employing default parameters, potentially leading to operational abnormalities. Consequently, it is highly recommended to update the following information to ensure the correct utilization of the ADCS-x in orbit. Users are advised to prepare this information in advance of receiving their ADCS-x to facilitate a seamless configuration process.

In this instruction, the frame of the ADCS is represented in B for short; the frame of the satellite is represented in H ; the frame of the CMG is represented in G ; and the frame of all sensors is represented in S for short.

Step 1: Configuration

Before initiating the satellite's operational phase, it is essential to configure all relevant parameters for optimal performance. Refer to Table 7-1 for a comprehensive list of parameters and their recommended settings. Carefully set each parameter according to the specified values to guarantee optimal performance during the mission.

Table 7-1. Parameters required for configuring the ADCS prior to simulation tests or space missions

Parameter	Description
M	Total mass of the satellite (unit: kg)
${}^H\mathbf{I}$	3x3 inertia matrix expressed in H frame (unit: kg-m ²)
${}^B\mathbf{r}_{H/B}$	3x1 position vector of the center of mass (COM) of the satellite relative to the ADCS in the ADCS frame (unit: m)
$\mathbf{q}_{H/B}$	The orientation of the satellite relative to ADCS
${}^B\mathbf{r}_{G/B}$	The 3x1 position vector of the CMG center relative to ADCS in ADCS frame (unit: m)
$\mathbf{q}_{B/S}$	The orientation of the ADCS relative to each sensors Tensor Tech will initially configure the placement of the six fine sun sensors perpendicular to each other as the default setting. Users are advised to review and adjust this configuration as necessary.
A unit vector in the Satellite Frame that	one 3x1 unit vector

Parameter	Description
designated the normal-line vector of the satellite's face equipping with deployed solar panels	

The parameters detailed in Table 7-1 remain non-adjustable during on-orbit operations, with exceptions made for the mass properties and the center of the CMG's position vector. Any expulsion of fuel from the thruster's fuel tank will result in alterations to the satellite's mass properties. This necessitates the provision of a mechanism enabling the On-Board Computer (OBC) to modify the ADCS's mass properties configuration in space.

Step 2: Commissioning

Detumbling Mode Activation

Once the satellite has been successfully deployed, it is necessary to transition it into detumbling mode. Execute the following steps:

- Utilize the OBC to control ADCS for detumbling mode.
- Verify that the satellite is in detumbling mode by checking register STAT.
- check registers RATEX, RATEY, and RATEZ to make sure that the tumbling rate is decreasing.

Time and Position Synchronization

To ensure accurate attitude determination and control, update the orbiting information to the ADCS with the following steps:

- Without GNSS:
 - Update register TIME, DATE and TLE
 - Confirm time and position is synchronized by reading registers TIME, DATE and LLA
- With GNSS:
 - Check the STAT.GNSS to check if the GNSS status is normal.
 - Check the value inside register TIME, DATE and LLA.

Sun Pointing Mode Activation

Before initiating any payload operations, it is crucial to ensure that the sun vector is valid.

Follow these steps to enter sun pointing mode:

- Verify the validity of the sun vector
- Initiate the sun pointing mode.

Solar Panel Deployment

Deploy the satellite's solar panels to harness solar energy for power. Follow these steps:

- Confirm the satellite is in a stable configuration
- Execute the command to deploy the solar panels through OBC
- Monitor telemetry data to verify the status of deployment

Step 3: Operational Phase

After the satellite's battery is sufficiently charged, set the ADCS system to the desired operational mode based on mission requirements. Execute the following steps:

- Confirm battery charge status
- Set ADCS to LVLH mode, Fine Pointing mode, or Target Tracking mode based on mission specifications
- Monitor telemetry data to ensure successful mode transition

Adhering to these procedures will contribute to the smooth and efficient execution of satellite operations, minimizing the risk of errors and ensuring the successful accomplishment of mission objectives.

REVISION HISTORY

Date	Editor	Version	Contents
2020.09.26	S. Lee	V1.0.0	Establishing initial parameters and definitions
2023.04.15	S. Lee	V2.0.0	Add ADCS-MTQ description
2023.07.01	J. Liao, S. Lee	V2.0.1	Update Ch5., Ch6. and Ch12.
2023.07.11	S. Lee	V2.0.2	Update Ch2.
2023.08.22	S. Lee	V2.0.3	Move Ch1~Ch5 to <i>ADCS Series Datasheet</i>
2023.09.10	S. Lee, W.Fu	V2.0.4	Update Ch2. Update ch7.2, SNID register
2023.09.27	T. Yen	V2.0.5	Revise the environmental verification specs in Ch.4.
2023.10.02	T. Yen	V2.0.6	Revise the format and the style of the documentation
2023.11.14	A. Huang	V2.0.7	Wiring diagram reference update
2023.11.28	J. Liao	V2.0.8	Update 6.3 and 6.5
2024.04.18	S. Lee, C. Hu, D. Hsieh, A. Huang,	V2.0.9	1. Remove all protection shroud as optional description. Protection shroud is a standard item for ADCS products. 2. Environmental Test Standard Section is seperated as individual document from user manual. 3. Electrical Interface and Mechanical Interface Detail is separated as ICD from user manual.

REFERENCES

#	Document Name	Version
[1]	3D CAD Model of ADCS-x in Configuration 1	V1.3
[2]	3D CAD Model of ADCS-x in Configuration 2	V1.3
[3]	3D CAD Model of FSS-15	V2.2
[4]	3D CAD Model of FSS-15M	V2.2
[5]	3D CAD Model of CMG-10m	V9.6
[6]	Tensor Tech CO., LTD. (2023). User Manual for CMG-10m CMG-10m Control Moment Gyro based on Spherical Motor Technology.	V2.4.2
[7]	Tensor Tech CO., LTD. (2023). User Manual for FSS-15 Nano Fine Sun Sensor.	V2.5.6
[8]	NASA Goddard Space Flight Center. (2021). GSFC-STD-7000B General Environmental Verification Standard (GEVS).	GSFC-STD-7000B
[9]	European Space Components Coordination. (2016). TOTAL DOSE STEADY-STATE IRRADIATION TEST METHOD. ESCC Basic Specification No. 22900.	Issue.5
[10]	IRRADIATION TEST METHOD. ESCC Basic Specification No. 22900.	Issue.5
[11]	John Sonnenberg, Raveon Technologies Corp. (2018). Technical Brief AN236 Rev A Serial Communications RS232, RS485, RS422.	Rev A